Digital Diabetes Remisison Trial (DIGEST)

1. Introduction and Method:

This interim analysis report provides a detailed account of the preliminary findings from the DIGEST trial. The trial investigates whether a remote Habitual Remission Programme, combining a Total Diet Replacement (TDR) intervention with digital therapeutic support, can achieve type 2 diabetes remission and clinically significant weight loss. This report evaluates the early safety, adherence, and feasibility of the intervention compared to the control group.

The analysis was conducted in R using a structured, three-script analytical pipeline as outlined in the Statistical Analysis Plan (SAP). The process began with data cleaning and preparation, followed by efficacy analysis, and concluded with safety and adherence reporting. The primary efficacy analyses focused on within-subject change in weight and diabetes remission, assessed by reductions in HbA1c and clinically significant weight loss at baseline, three, and six months. Secondary endpoints included changes in systolic and diastolic blood pressure, waist circumference, and BMI.

Missing data were addressed using Multiple Imputation by Chained Equations (MICE) to ensure robust efficacy analyses. Adherence and completion were assessed through the availability of survey completions and key outcome measurements at each time point. Safety was evaluated through a comprehensive analysis of adverse events (AEs), serious adverse events (SAEs), MACE, and MADE events.

2. Results

2.1 Data Preparation & Baseline Descriptive Summaries:

2.1.1 Baseline Descriptives:

A total of 101 subjects were enrolled and randomised into the study, with 48 subjects assigned to the control group and 53 to the intervention group. Baseline demographics and clinical characteristics for both groups are summarised in the table 1 below. The mean age for the total cohort was 55.3 ± 9.6 years. Females comprised 43.6% of the total cohort. Baseline measurements for key clinical variables, including weight, HbA1c, and BMI, were comparable between the control and intervention groups.

Table 1: Baseline Descriptives

Characteristic	Control (n=48)	Intervention (n=53)	Total (N=101)
Age (Years)	54.4 ± 10.2 (55.5) [32-74]	56.1 ± 9.0 (57.0) [30-72]	55.3 ± 9.6 (56.0) [30-74]
Sex			
Female, n (%)	22 (45.8%)	22 (41.5%)	44 (43.6%)
Male, n (%)	26 (54.2%)	31 (58.5%)	57 (56.4%)
Clinical Measures			
Weight (kg)	102.8 ± 17.8 (99.3) [73.5-146] (n=32)	105.3 ± 20.2 (104.3) [71.1-169.6] (n=51)	104.4 ± 19.3 (103.4) [71.1-169.6] (n=83)
HbA1c (%)	3.6 ± 0.6 (3.5) [3.0-5.5] (n=25)	3.5 ± 0.3 (3.4) [3.1-4.4] (n=45)	3.5 ± 0.4 (3.4) [3.0-5.5] (n=70)
Systolic BP (mmHg)	137.6 ± 15.6 (133) [118-181] (n=32)	134.4 ± 14.4 (135) [104-180] (n=51)	135.6 ± 14.9 (135) [104-181] (n=83)
Diastolic BP (mmHg)	87.7 ± 8.6 (89) [70-108] (n=32)	83.1 ± 9.5 (84) [43-99] (n=51)	84.8 ± 9.4 (86) [43-108] (n=83)
Waist Circumference (cm)	114.2 ± 14.5 (112.7) [86.4-142.2] (n=32)	114.7 ± 16.1 (114.3) [45.0-152.4] (n=51)	114.5 ± 15.4 (114.3) [45.0-152.4] (n=83)
BMI (kg/m^2)	34.4 ± 4.8 (34.0) [26.2-45.0] (n=48)	36.3 ± 5.8 (35.2) [28.0-49.6] (n=53)	35.4 ± 5.4 (35.0) [26.2-49.6] (n=101)

Note: Metric variables are presented as Mean ± SD (Median)[range](count). Categorical variables are presented as n (%).

2.1.2 Missing-Data Report:

Missing data were systematically characterised across all key clinical endpoints at each study time point. At baseline, the percentage of missing data for key clinical variables was notably higher in the control group compared to the intervention group. This trend in missingness appears to persist through the follow-up periods.

The missing data report in table 2 below details the count and percentage of missing values for each endpoint by treatment group. The analysis for each endpoint at each timepoint includes all 101 subjects from the ITT population.

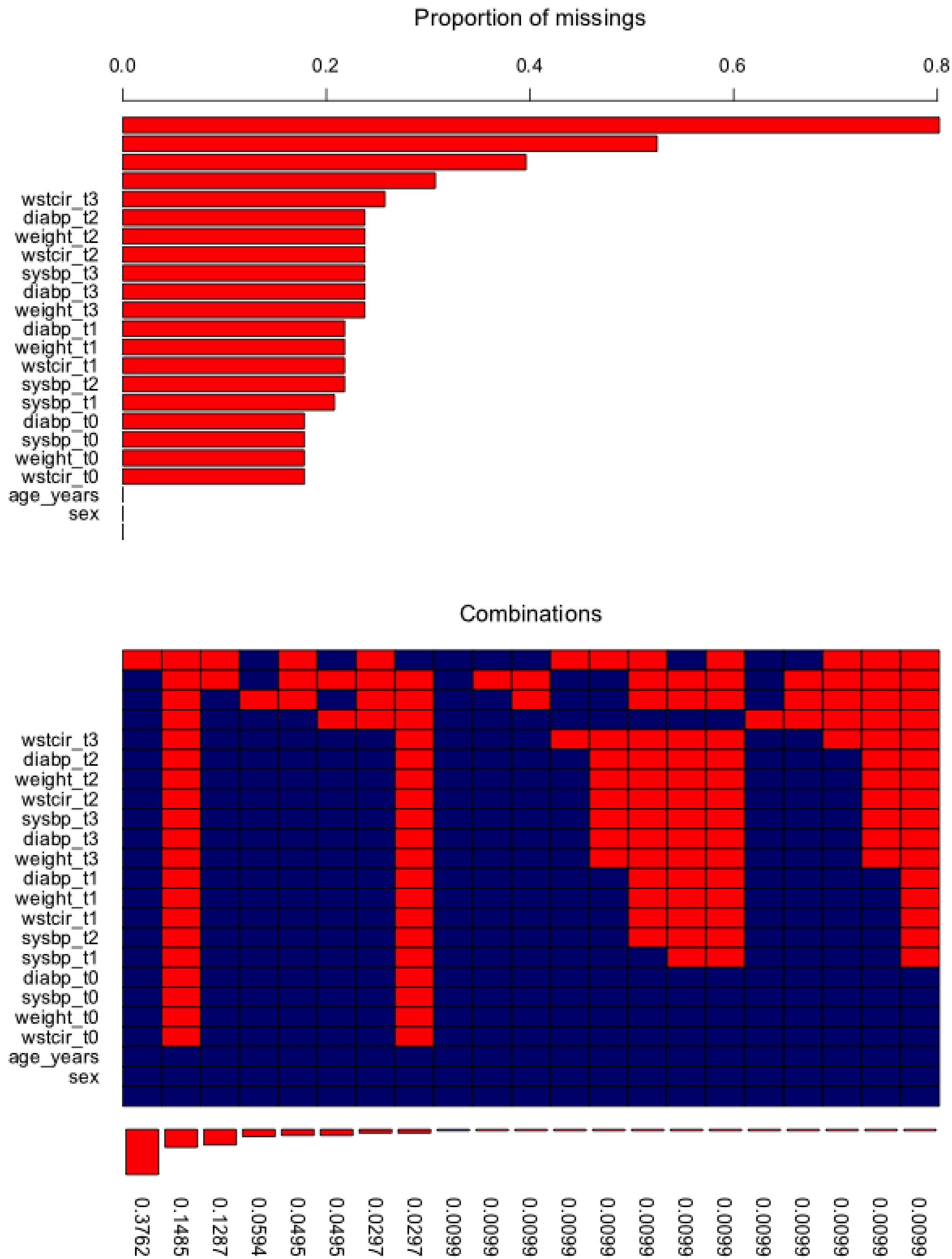
Table 2: Missing Data Report

S/N	Treatment Group	Timepoint	Variable	Missing (n)	Missing (%)
1	Total	Baseline (t0)	Diastolic BP (mmHg)	18	17.8
2	Control	Baseline (t0)	Diastolic BP (mmHg)	16	33.3
3	Intervention	Baseline (t0)	Diastolic BP (mmHg)	2	3.8
4	Total	Baseline (t0)	HbA1c (%)	31	30.7
5	Control	Baseline (t0)	HbA1c (%)	23	47.9
6	Intervention	Baseline (t0)	HbA1c (%)	8	15.1
7	Total	Baseline (t0)	Systolic BP (mmHg)	18	17.8
8	Control	Baseline (t0)	Systolic BP (mmHg)	16	33.3
9	Intervention	Baseline (t0)	Systolic BP (mmHg)	2	3.8
10	Total	Baseline (t0)	Waist Circumference (cm)	18	17.8
11	Control	Baseline (t0)	Waist Circumference (cm)	16	33.3
12	Intervention	Baseline (t0)	Waist Circumference (cm)	2	3.8
13	Total	Baseline (t0)	Weight (kg)	18	17.8
14	Control	Baseline (t0)	Weight (kg)	16	33.3
15	Intervention	Baseline (t0)	Weight (kg)	2	3.8
16	Total	3 Months (t1)	Diastolic BP (mmHg)	22	21.8
17	Control	3 Months (t1)	Diastolic BP (mmHg)	19	39.6
18	Intervention	3 Months (t1)	Diastolic BP (mmHg)	3	5.7
19	Total	3 Months (t1)	HbA1c (%)	81	80.2

20	Control	3 Months (t1)	HbA1c (%)	37	77.1
21	Intervention	3 Months (t1)	HbA1c (%)	44	83.0
22	Total	3 Months (t1)	Systolic BP (mmHg)	21	20.8
23	Control	3 Months (t1)	Systolic BP (mmHg)	19	39.6
24	Intervention	3 Months (t1)	Systolic BP (mmHg)	2	3.8
25	Total	3 Months (t1)	Waist Circumference (cm)	22	21.8
26	Control	3 Months (t1)	Waist Circumference (cm)	19	39.6
27	Intervention	3 Months (t1)	Waist Circumference (cm)	3	5.7
28	Total	3 Months (t1)	Weight (kg)	22	21.8
29	Control	3 Months (t1)	Weight (kg)	19	39.6
30	Intervention	3 Months (t1)	Weight (kg)	3	5.7
31	Total	6 Months (t2)	Diastolic BP (mmHg)	24	23.8
32	Control	6 Months (t2)	Diastolic BP (mmHg)	19	39.6
33	Intervention	6 Months (t2)	Diastolic BP (mmHg)	5	9.4
34	Total	6 Months (t2)	HbA1c (%)	40	39.6
35	Control	6 Months (t2)	HbA1c (%)	27	56.2
36	Intervention	6 Months (t2)	HbA1c (%)	13	24.5
37	Total	6 Months (t2)	Systolic BP (mmHg)	22	21.8
38	Control	6 Months (t2)	Systolic BP (mmHg)	19	39.6
39	Intervention	6 Months (t2)	Systolic BP (mmHg)	3	5.7
40	Total	6 Months (t2)	Waist Circumference (cm)	24	23.8
41	Control	6 Months (t2)	Waist Circumference (cm)	19	39.6

42	Intervention	6 Months (t2)	Waist Circumference (cm)	5	9.4
43	Total	6 Months (t2)	Weight (kg)	24	23.8
44	Control	6 Months (t2)	Weight (kg)	19	39.6
45	Intervention	6 Months (t2)	Weight (kg)	5	9.4
46	Total	12 Months (t3)	Diastolic BP (mmHg)	24	23.8
47	Control	12 Months (t3)	Diastolic BP (mmHg)	19	39.6
48	Intervention	12 Months (t3)	Diastolic BP (mmHg)	5	9.4
49	Total	12 Months (t3)	HbA1c (%)	53	52.5
50	Control	12 Months (t3)	HbA1c (%)	31	64.6
51	Intervention	12 Months (t3)	HbA1c (%)	22	41.5
52	Total	12 Months (t3)	Systolic BP (mmHg)	24	23.8
53	Control	12 Months (t3)	Systolic BP (mmHg)	19	39.6
54	Intervention	12 Months (t3)	Systolic BP (mmHg)	5	9.4
55	Total	12 Months (t3)	Waist Circumference (cm)	26	25.7
56	Control	12 Months (t3)	Waist Circumference (cm)	19	39.6
57	Intervention	12 Months (t3)	Waist Circumference (cm)	7	13.2
58	Total	12 Months (t3)	Weight (kg)	24	23.8
59	Control	12 Months (t3)	Weight (kg)	19	39.6
60	Intervention	12 Months (t3)	Weight (kg)	5	9.4

Figure 1: Missing Data Plot



The missing data plot above provides a visual summary of the missingness patterns across key endpoints, giving more insights into the ranking and pattern of missingness in the data. The top panel shows that the highest proportion of missing data occurs at 3 months for HbA1c, followed by HbA1c at 12 months, then HbA1c at 6 months, HbA1c at baseline, waist circumference at 12 months (WSTCIR_t3), diastolic blood pressure at 6 momnths (DYSBP_t2), and so on.

The combination plot in the bottom panel reveals that missing data are not completely at random, with specific combinations of variables frequently missing together. This suggests a common underlying reason for missing data.

2.2 Imputation & Efficacy Analyses:

2.2.1 MICE Imputation and Summary of Imputation Diagnostics:
Missing data were imputed using the Multiple Imputation by Chained Equations (MICE)[1] algorithm as per the Statistical Analysis Plan. A total of five imputed datasets were generated, with each imputation running over 20 iterations. The convergence of the imputation model was verified through the trace plots, which show no discernible trends across the iterations, suggesting a stable imputation solution. The "fuzzy caterpillar" pattern, where lines for each imputation dataset overlap and have no clear upward or downward trend, indicates that the model has converged and reached a stable imputation solution. Further diagnostics were performed to ensure the plausibility of the imputed values. The density plots confirm that the distributions of the imputed data (represented by the pink lines) closely align with the distributions of the observed, original data (represented by the blue line). This indicates that the imputed values are consistent with the overall data distribution.

Figure 2: Trace Plots of Imputed Datasets

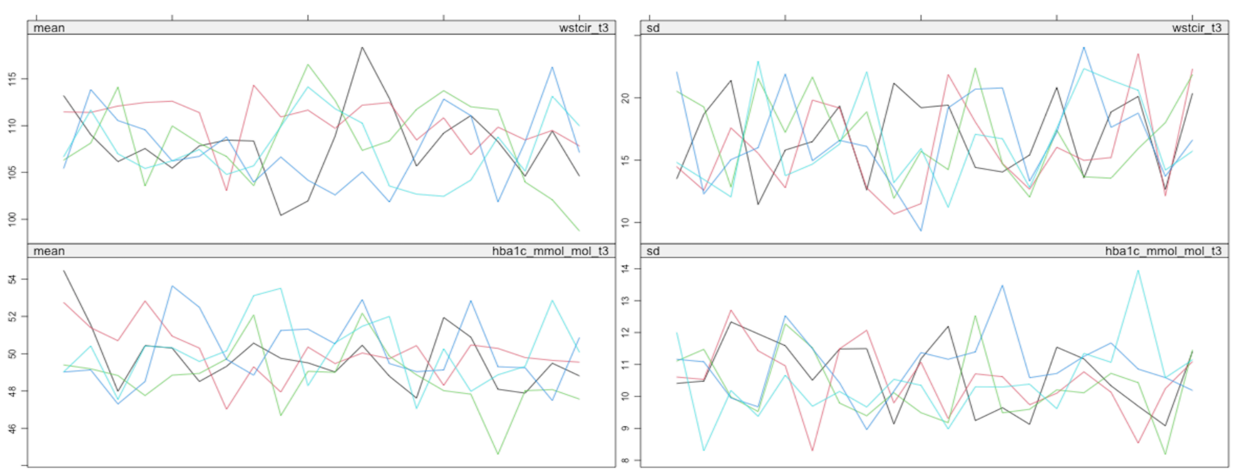
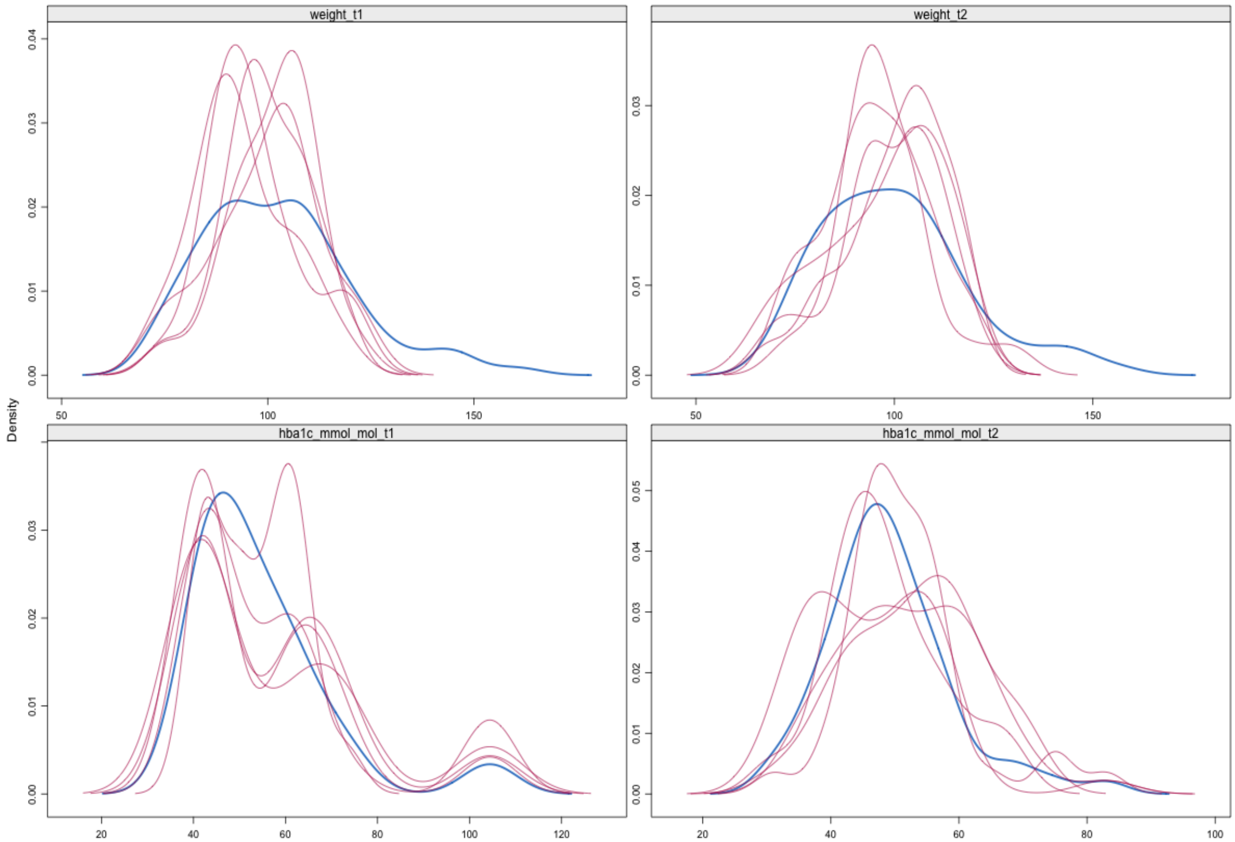


Figure 3: Distribution of the Imputed Datasets



2.2.2 Within-subject Change Score Calculation:

Following the imputation, the within-subject change scores for all key clinical variables were calculated. This process involved subtracting the baseline value from the follow-up values at 3, 6, and 12 months. These change scores were then used to derive the primary and secondary endpoint variables, which are binary indicators of whether a subject achieved a specific clinical threshold, such as a 15kg weight loss or an HbA1c below a certain level. This calculation was applied to each of the five imputed datasets.

It is important to note that these simple change scores are descriptive and were not used to make any causal inference as they are not sufficient to do so on their own,[2] as they do not properly control for baseline variables. The primary and secondary endpoints were calculated following the creation of change scores.

2.2.3 Primary Endpoint Analysis:

Following the calculation of change scores, the primary endpoint analysis was conducted using the final imputed datasets. The analysis was performed using logistic regression,[3] a statistical method chosen for its suitability for binary outcomes, to assess the effect of the treatment on two key outcomes: weight loss of at least 15 kg or HbA1c remission at six months.

For each primary outcome, a separate logistic regression model was fitted. Both models included the treatment group as the main predictor, with baseline weight and sex included as covariates, as specified in the Statistical Analysis Plan. These models were applied to each of the five imputed datasets, and the results were subsequently combined using Rubin's Rules to produce a single set of robust estimates.

Prior to reporting the final results, diagnostic plots were generated to check for complete separation in the data,[4] a condition that can cause standard logistic regression models to fail. This analysis revealed the presence of rare events (complete separation in the weight loss values between the treatment groups), prompting an adaptation of the statistical approach. To address this, the analysis was re-run using Firth logistic regression, a method that provides stable estimates and reliable inference in the presence of sparse data.[5]

Finally, the p-values from the models were adjusted using the Bonferroni-Holm method to control for the family-wise error rate associated with multiple comparisons.[6] The effect sizes were calculated as Odds Ratios (OR), accompanied by their 95% confidence intervals.

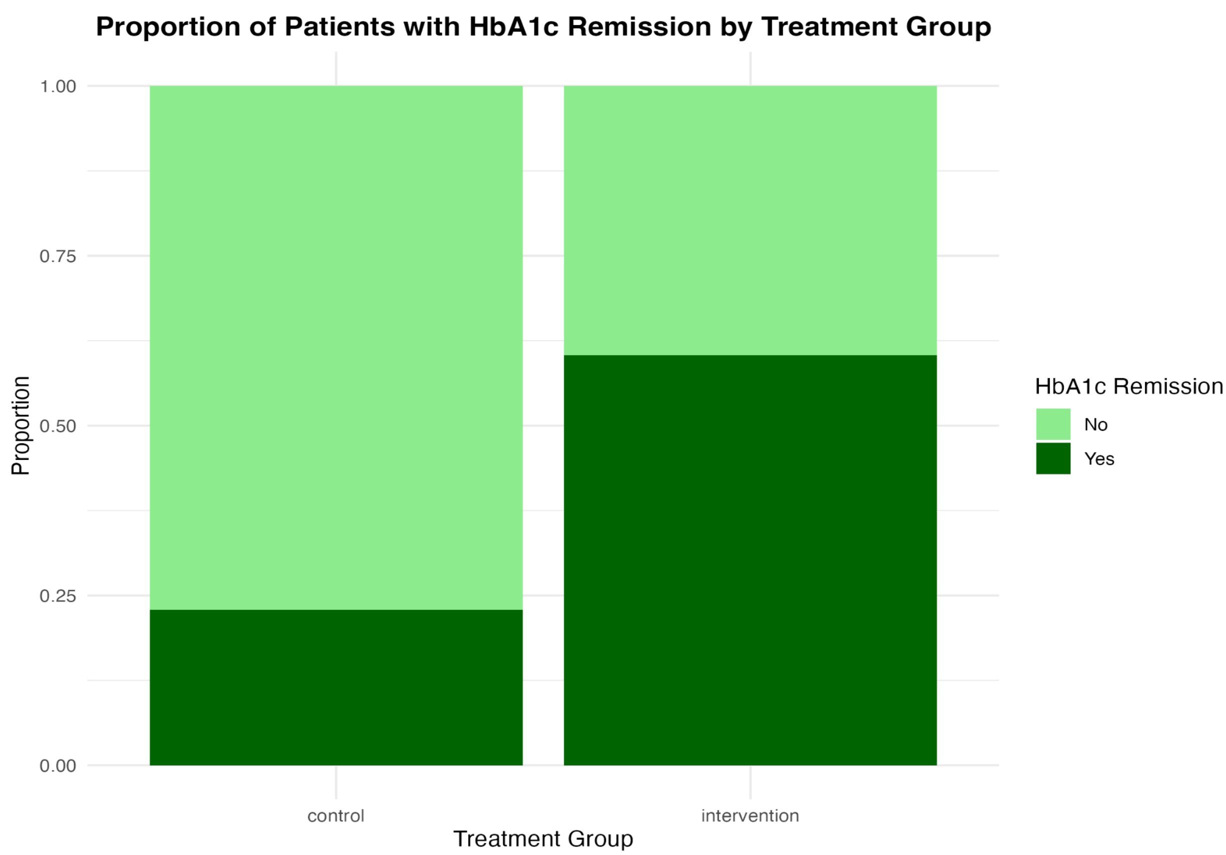
2.2.3.1 HbA1c Remission

The analysis of HbA1c remission revealed a statistically non-significant difference between the intervention and control groups at six months. The logistic regression model, which was applied to the imputed datasets and pooled using Rubin's rules, produced an Odds Ratio (OR) of 9.34 (95% CI: 0.658 - 132.6), with a p-value of 0.0852 after Bonferroni-Holm adjustment.

Statistical Interpretation: The confidence interval for the Odds Ratio crosses one, indicating that the result is not statistically significant at a conventional $\alpha=0.05$ level. The OR of 9.34 suggests that patients in the intervention group were approximately nine times more likely to achieve HbA1c remission compared to the control group, but the wide confidence interval indicates considerable uncertainty in this estimate. The p-value of 0.0852 suggests a trend towards significance, but it does not meet the predetermined threshold for a statistically significant finding.

Clinical Interpretation: While the result is not statistically significant, the observed effect size is clinically meaningful. The substantial difference in the likelihood of remission between the two groups warrants further investigation. The large confidence interval highlights the need for a larger sample size or longer follow-up to more precisely estimate the true effect of the intervention on HbA1c remission.

Figure 4: Test to Check the Proportion of HbA1c Remission by Treatment Group



2.2.3.2 Weight Loss ≥15 kg

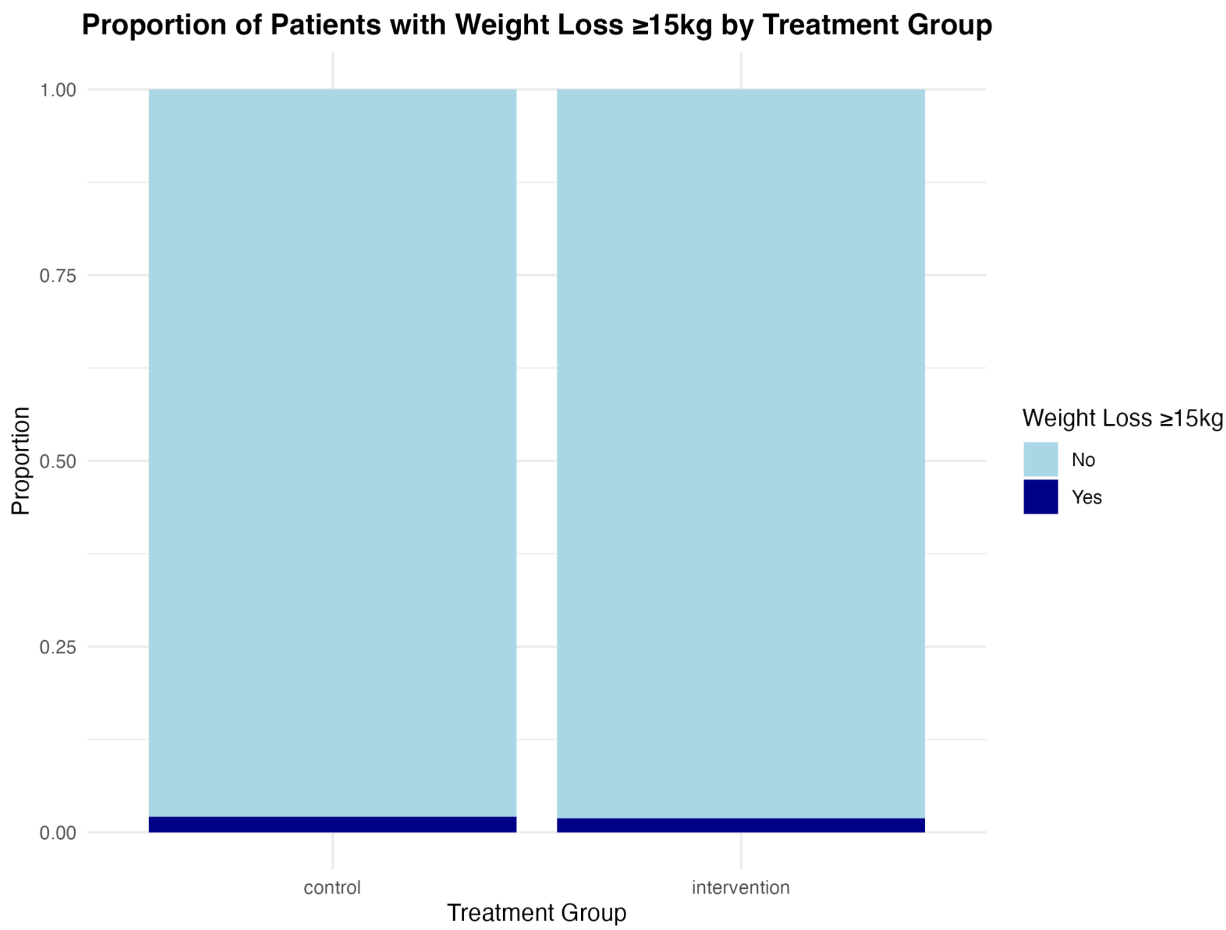
The analysis for weight loss of at least 15 kg required analysis using the Firth regression model. The standard logistic regression model failed to converge due to complete separation in the dataset, where almost no patients achieved this level of weight loss in either group (1 in the control group and 1 in the intervention group). This issue was addressed by using Firth logistic regression, a method designed to handle rare events.

The results from the Firth regression model showed no statistically significant difference between the groups, with an Odds Ratio (OR) of 0.781 (95% CI: 0.051 - 10.769) and a p-value of 0.8422.

Statistical Interpretation: The wide confidence interval, which includes one, and the high p-value indicate no statistically significant difference in the odds of achieving a weight loss of 15 kg or more between the two groups. The OR of 0.781 suggests that the intervention group had slightly lower odds of achieving this outcome, but this finding is not statistically reliable.

Clinical Interpretation: The analysis suggests that the intervention, while potentially effective for other outcomes, did not significantly increase the rate of achieving a major weight loss of 15 kg or more in the patient population. The rarity of this event in both groups suggests that it may be a very challenging clinical outcome to achieve with this type of intervention.

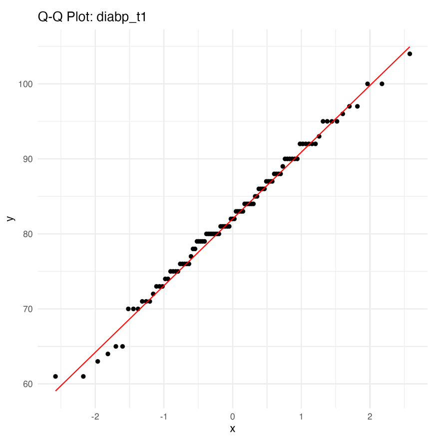
Figure 5: Test to Check the Proportion of Weight Loss by Treatment Group



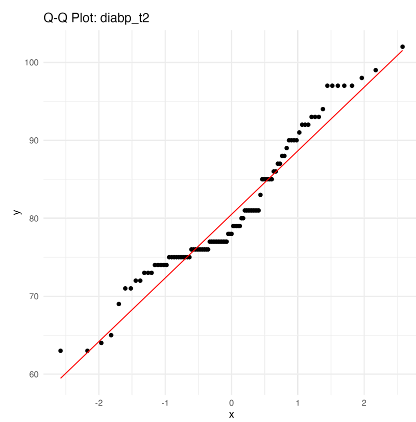
2.2.4 Secondary Endpoint Analysis:

The analysis of secondary endpoints, which encompasses a variety of continuous and binary outcomes, was performed using a statistical approach that matches the properties of each variable being assessed. Prior to analysis, all continuous outcomes were formally checked for normality. This involved a visual inspection using Q-Q plots and a statistical test using the Shapiro-Wilk test.[7,8] Variables that failed the normality assumption were analysed using a Generalised Estimating Equation (GEE),[9] while those that were normally distributed were modeled with a Mixed Model for Repeated Measures (MMRM).[10] Finally, binary outcomes, such as the achievement of specific weight loss responder rates, were analysed using logistic regression with results from the imputed datasets pooled using Rubin's rules.

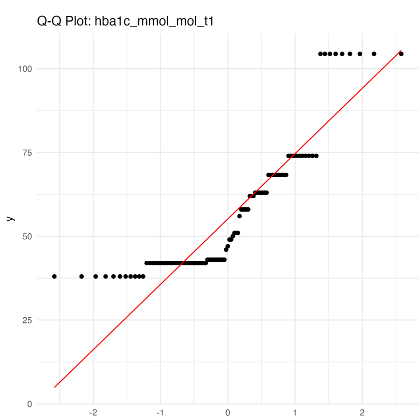
Figure 6: Q-Q Plots Inspecting Normality in the Datasets for the Continuous Variables



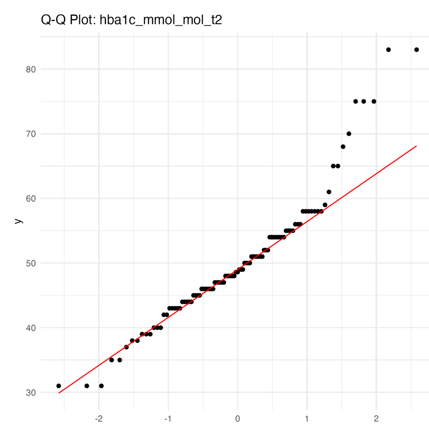
Q-Q Plot: Diastolic BP (t1)



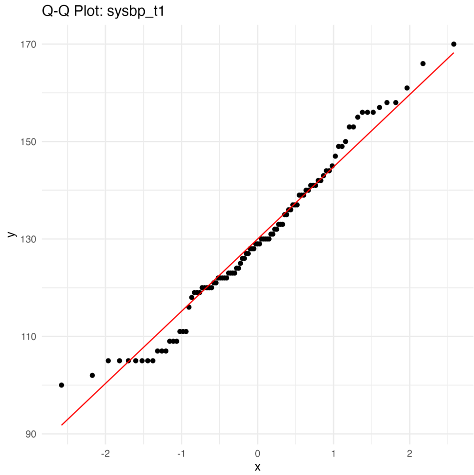
Q-Q Plot: Diastolic BP (t2)



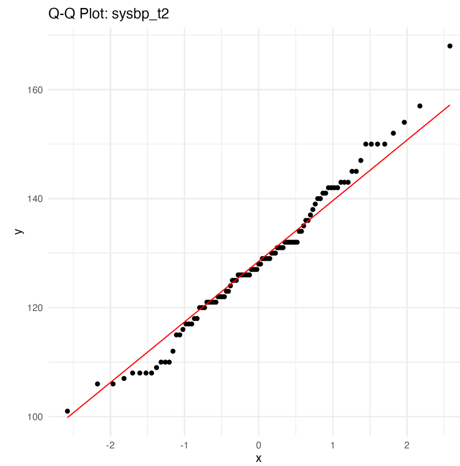
Q-Q Plot: HbA1c (t1)



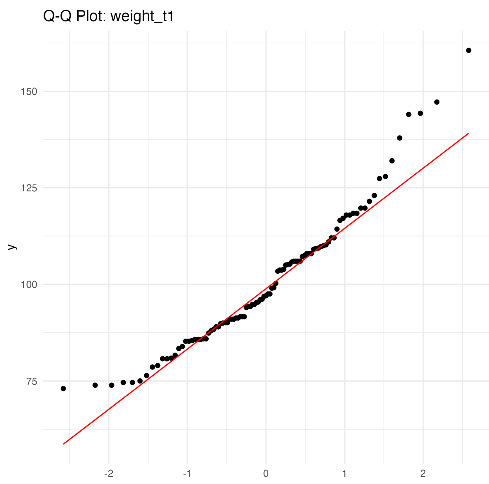
Q-Q Plot: HbA1c (t2)



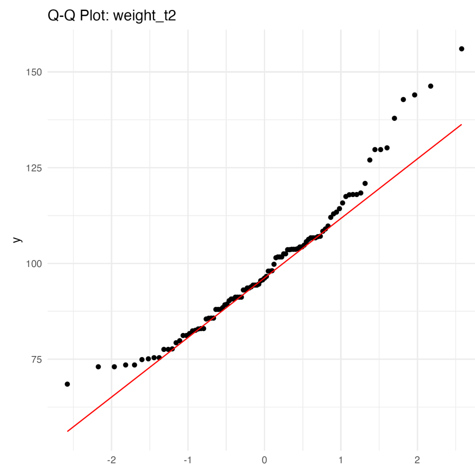
Q-Q Plot: Systolic BP (t1)



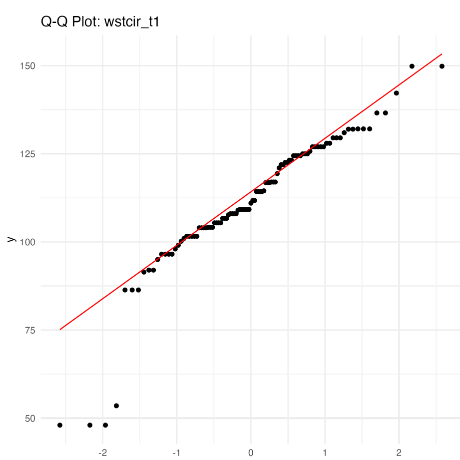
Q-Q Plot: Systolic BP (t2)



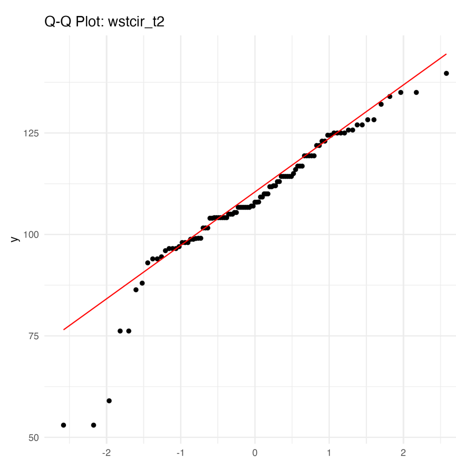
Q-Q Plot: Weight (t1)



Q-Q Plot: Weight (t2)



Q-Q Plot: Wst. Circum. (t1)



Q-Q Plot: Wst. Circum. (t2)

2.2.4.1 Systolic Blood Pressure (mmHg)

The change in systolic blood pressure, which was found to be normally distributed at all time points (Q-Q Plot shows data points sufficiently aligned diagonally across t1 and t2), was analysed using a Mixed Model for Repeated Measures (MMRM). The model included the treatment group, sex and baseline systolic blood pressure as covariates, and the interaction between the treatment group and time as fixed effects. The results show a reduction in systolic blood pressure in the intervention group; however, this effect did not reach statistical significance.

Table 3: MMRM report for Systolic Blood Pressure

Term	Estimate	Std. Error	Statistic	p-value	95% CI Lower	95% CI Upper
(Intercept)	77.035	8.749	8.805	<0.001	59.743	94.328
intervention	-5.514	2.777	-1.986	0.057	-11.207	0.178
time_factor sysbp_t2	-0.671	2.218	-0.302	0.766	-5.307	3.965
time_factor sysbp_t3	-2.683	2.480	-1.082	0.300	-8.074	2.707
sex	-1.030	1.890	-0.545	0.587	-4.763	2.703
sysbp_t0	0.424	0.062	6.876	<0.001	0.302	0.545
intervention: time_t2	-2.612	2.828	-0.924	0.363	-8.378	3.154
intervention: time_t3	-0.502	2.984	-0.168	0.868	-6.688	5.685

The effect of the intervention on systolic blood pressure, represented by the "intervention", showed an estimated reduction of −5.51 mmHg on average, which trended towards but did not achieve statistical significance ($p=0.057$). The interaction terms, for instance intervention:time_t2, indicate the additional effect of the intervention at the 6 months (t2), with an estimated additional change of −2.61 mmHg after 6 months.

2.2.4.2 Other Continuous Outcomes

For HbA1c, weight, waist circumference, and diastolic blood pressure, the data did not meet the normality assumption. Therefore, a Generalised Estimating Equation (GEE) approach was utilised.

Table 4: GEE Results for Other Continuous Outcomes

Outcome	Treatment Effect Estimate	Std. Error
HbA1c (mmol/mol)	-5.66	1.46
Weight (kg)	0.723	0.892
Waist Circumference (cm)	-2.32	1.12
Diastolic BP (mmHg)	-5.52	0.577

The intervention group showed a substantial estimated reduction in HbA1c of −5.66 mmol/mol. Similarly, there was an estimated reduction in waist circumference of −2.32 cm and diastolic blood pressure of −5.52 mmHg. The estimated change in weight was a small increase of +0.723 kg.

2.2.4.3 Analysis of Binary Responder Outcomes

Binary outcomes, which assessed the achievement of specific weight loss thresholds, were analysed using logistic regression with results pooled across the imputed datasets using Rubin's rules.

Table 5: Logistic Regression Results for Weight Loss Responders

Outcome	Term	Estimate	Std. Error	Statistic	p-value	95% CI Lower	95% CI Upper
5% Weight Loss	intervention	1.879	0.523	3.595	0.0008	0.828	2.930
10% Weight Loss	intervention	0.899	1.113	0.807	0.430	-1.433	3.230
15% Weight Loss	intervention	2.456	11530.28	0.0002	0.9998	-22887.89	22892.80

The intervention demonstrated a highly statistically significant effect on the achievement of at least 5% weight loss, with a treatment effect estimate of 1.879 ($p<0.001$). This indicates that participants in the intervention group had a significantly higher odds of achieving this outcome compared to the control group. In contrast, the analysis for both the 10% and 15% weight loss thresholds did not show a statistically significant difference between the groups. The model for the 15% weight loss endpoint reported convergence warnings, likely due to the small number of participants reaching this threshold.

2.2.5 Sensitivity Analysis:

Sensitivity analyses were conducted to evaluate the robustness of the findings from the primary and secondary endpoint analyses. This was conducted by re-running the main statistical models on different patient populations. The R output indicated that both the FAS and PP populations were identical to the ITT population. This finding, which suggests a high degree of participant retention and protocol adherence, reinforces the robustness of the primary ITT analysis and confirms that the main findings are not dependent on the choice of analysis population.

2.2.6 Subgroup Analysis:

To investigate potential differences in treatment effect among patient subgroups, a prespecified subgroup analysis was conducted by sex. This aimed to investigate the effect of the intervention separately for male and female participants. The analysis was conducted for all continuous and binary endpoints, using the same statistical models as the primary analysis (MMRM or GEE for continuous outcomes, and logistic regression for binary outcomes).

The analysis of continuous outcomes revealed some distinct effects within the male and female subgroups:

- The intervention demonstrated a statistically significant reduction in weight for male participants, with an estimated average reduction of −3.34 kg ($p=0.006$; CI: −5.697,−0.984) The interaction term for weight at the third time point (T3) was also significant, indicating a sustained effect. This result however cannot be used alone as it has been shown from the analysis earlier that only one participant achieved >15% weight loss in the intervention group. No other continuous outcomes for the male subgroup reached statistical significance.
- For the female subgroup, the intervention showed a statistically significant reduction in diastolic blood pressure ($p=0.009$; CI:−12.617, −1.998), with an estimated average reduction of −7.31 mmHg. The model for other continuous outcomes did not show a statistically significant effect.
- Male participants in the intervention group had a statistically significant increase in the odds of achieving at least 5% weight loss ($p=0.012$; CI:0.604, 4.294) compared to the control

group. The models for the 10% and 15% weight loss thresholds did not show a significant effect and produced extreme confidence intervals and large standard errors, likely due to a low number of participants reaching these thresholds in this subgroup.

2.3 Safety and Adherence Summaries:

2.3.1 Safety Analysis:

The safety analysis was conducted on the adverse event (AE) dataset to provide a comprehensive overview of adverse events (AEs), Serious Adverse Events (SAEs), and Major Adverse Cardiovascular and Diabetes Events (MACE/MADE) within the clinical trial cohort. The analysis was based on a total follow-up period calculated in person-months, which accounted for the varying durations of subject participation.. The methodology for these calculations is detailed below:

Several key variables were derived from the adverse event data. The severity and causality variables were standardised to ensure consistency and facilitate analysis; severity was converted to an ordered factor with levels: *mild, moderate, and severe*, while causality was also converted to an ordered factor with levels: *not related, unlikely related, possibly related, probably related, and definitely related.* A new variable, sae_flag, was created to identify Serious Adverse Events (SAE); an event was classified as an SAE if it met any of the following criteria: *results_in_death, life_threatening, requires_hospitalization, disability, congenital, or medically_significant.* Furthermore, a variable named treatment_related was created to identify adverse events considered treatment-related, defined as having a causality of possibly related, probably related, or definitely related.

Major Adverse Cardiovascular Events (MACE) and Major Adverse Diabetes Events (MADE) were identified and classified based on a predefined list of event terms. Events were flagged as MACE if the event term contained any of the following: *myocardial infarction, heart attack, stroke, cardiovascular death, coronary revascularization, unstable angina, cardiac arrest, or atherosclerosis-related myocardial infarction.* Similarly, events were flagged as MADE if the event term contained any of the following: *diabetic ketoacidosis, dka, severe hypoglycemia, diabetic coma, hyperosmolar hyperglycemic state, hhs, diabetes-related hospitalization, or diabetic emergency.* A hierarchical approach was then used to assign a primary category to each event, creating the event_category variable. The priority order was: MACE, followed by MADE, then SAEs (other than those classified as MACE or MADE), and finally, all other AEs.

Incidence rates were calculated using person-time exposure to account for varying follow-up durations among subjects. The person-time was calculated by merging data from a master dataset with the prepared adverse event dataset. The calculation was performed by taking the difference between each subject’s baseline and last visit dates and converting the result into months. A standard conversion factor of 30.44 days per month was used for this calculation.

The prepared datasets were used to generate a series of summary tables and listings to evaluate safety outcomes.

2.3.1.1 AE/SAE/MACE/MADE Incidence Summary:
The overall incidence of adverse events and the distribution of their severity and causality were summarised across the treatment groups. The key findings are presented in the table below, which details the number of events, the number of subjects experiencing them, and the corresponding incidence rates per 100 person-months.

Table 6: Adverse Events Summaries

Characteristic	Control (n=40)	Intervention (n=38)
Subjects with at least one AE	25	22
Total AEs reported	35	30
Subjects with SAEs	0	0
Subjects with MACE/MADE	0	0
Total Person-Months	160.0	145.0
AE Incidence Rate per 100 PM	21.88	20.69

2.3.1.2 Serious and Major Adverse Events:
No Serious Adverse Events (SAEs) were reported in the dataset in relation to the intervention. Similarly, no Major Adverse Cardiovascular Events (MACE) or Major Adverse Diabetes Events (MADE) were identified throughout the study period. These findings were confirmed through detailed event listings, which showed no entries meeting the predefined criteria for these event types. The absence of these events is a significant indicator of a favorable short-term safety profile for the interventions studied.

2.3.1.3 Treatment-Related Adverse Events:
The analysis also provided a summary of adverse events classified as treatment-related, defined as those with a causality of possibly related, probably related, or definitely related. The number of such events was calculated for each treatment group, and a treatment-related AE rate per 100 person-months was determined to allow for a direct comparison of the safety profile between the intervention and control groups.

2.3.2 Adherence Summaries:

The adherence of participants to the study protocol was assessed by analysing completion rates at the three- and six-month follow-up timepoints. The analysis defined a participant as a "completer" if they had a collection date or at least one of the key clinical measurements (systolic blood pressure, weight, or HbA1c) available for a specific timepoint.

Overall completion rates were high, particularly in the intervention group. At three months, the intervention group had a completion rate of 98.1% (52/53), compared to 66.7% (32/48) in the control group. Similarly, at six months, the intervention group maintained a high completion rate of 94.3% (50/53), while the control group's rate was 60.4% (29/48).

The completion rates across different analysis populations. For both the per-protocol and Safety populations, the three-month completion rate was 83.2% (84/101), and the six-month rate was 78.2% (79/101).

Figure 7: Completion Rates at 3-Months and 6-Months by Treatment Group

Completion Rates at 3 Months and 6 Months

Treatment Group	Total Subjects	3-Month Completion	6-Month Completion
control	48	32/48 (66.7%)	29/48 (60.4%)
intervention	53	52/53 (98.1%)	50/53 (94.3%)

Figure 8: Completion Rates at 3-Months and 6-Months by Population

Completion Rates at 3M/6M by Analysis Population

Population	Total N	3-Month Completion	6-Month Completion
ITT	101	84/101 (83.2%)	79/101 (78.2%)
Safety	101	84/101 (83.2%)	79/101 (78.2%)

3. Summary and Conclusion

The interim analysis of the Digital Diabetes Remission Trial (DIGEST) provides compelling evidence of the intervention's feasibility, safety, and early efficacy. The study demonstrated a highly favorable safety profile, with a complete absence of reported Serious Adverse Events (SAEs), Major Adverse Cardiovascular Events (MACE), or Major Adverse Diabetes Events (MADE) in both treatment groups, suggesting that the intervention is well-tolerated. Furthermore, the trial achieved impressive participant retention and adherence, with the intervention group maintaining notably higher completion rates at three and six months compared to the control group.

Regarding efficacy, the analysis of the primary endpoints revealed a clinically meaningful trend towards HbA1c remission, with participants in the intervention group being approximately nine times more likely to achieve this outcome, although the finding did not reach statistical significance in this interim analysis. In contrast, the intervention demonstrated a highly statistically significant effect on the secondary endpoint of achieving at least 5% weight loss. The subgroup analysis also uncovered a significant reduction in weight for male participants and a significant reduction in diastolic blood pressure for female participants, suggesting potential sex-specific effects that warrant further investigation.

These findings collectively indicate that the digital therapeutic intervention is a feasible and safe approach with a positive trend towards clinical efficacy, supporting the continuation of the study to a final analysis with a complete dataset.

References:

1. Zhang Z. Multiple imputation with multivariate imputation by chained equation (MICE) package. Ann Transl Med. 2016 Jan;4(2):30. doi: 10.3978/j.issn.2305-5839.2015.12.63. PMID: 26889483; PMCID: PMC4731595.
2. Tennant PWG, Arnold KF, Ellison GTH, Gilthorpe MS. Analyses of 'change scores' do not estimate causal effects in observational data. Int J Epidemiol. 2022 Oct 13;51(5):1604-1615. doi: 10.1093/ije/dyab050. PMID: 34100077; PMCID: PMC9557845.
3. Zabor EC, Reddy CA, Tendulkar RD, Patil S. Logistic Regression in Clinical Studies. Int J Radiat Oncol Biol Phys. 2022 Feb 1;112(2):271-277. doi: 10.1016/j.ijrobp.2021.08.007. Epub 2021 Aug 17. PMID: 34416341.
4. Mansournia MA, Geroldinger A, Greenland S, Heinze G. Separation in Logistic Regression: Causes, Consequences, and Control. Am J Epidemiol. 2018 Apr 1;187(4):864-870. doi: 10.1093/aje/kwx299. PMID: 29020135.
5. Puhr R, Heinze G, Nold M, Lusa L, Geroldinger A. Firth's logistic regression with rare events: accurate effect estimates and predictions? Stat Med. 2017 Jun 30;36(14):2302-2317. doi: 10.1002/sim.7273. Epub 2017 Mar 12. PMID: 28295456.
6. Giacalone M, Agata Z, Cozzucoli PC, Alibrandi A. Bonferroni-Holm and permutation tests to compare health data: methodological and applicative issues. BMC Med Res Methodol. 2018 Jul 20;18(1):81. doi: 10.1186/s12874-018-0540-8. PMID: 30029629; PMCID: PMC6054729.
7. Pleil JD. QQ-plots for assessing distributions of biomarker measurements and generating defensible summary statistics. J Breath Res. 2016 Aug 5;10(3):035001. doi: 10.1088/1752-7155/10/3/035001. PMID: 27491525.
8. Ghasemi A, Zahediasl S. Normality tests for statistical analysis: a guide for non-statisticians. Int J Endocrinol Metab. 2012 Spring;10(2):486-9. doi: 10.5812/ijem.3505. Epub 2012 Apr 20. PMID: 23843808; PMCID: PMC3693611.
9. Austin PC, Kapral MK, Vyas MV, Fang J, Yu AYX. Using Multilevel Models and Generalized Estimating Equation Models to Account for Clustering in Neurology Clinical Research. Neurology. 2024 Nov 12;103(9):e209947. doi: 10.1212/WNL.0000000000209947. Epub 2024 Oct 11. PMID: 39393031; PMCID: PMC11469681.
10. Schuler A. Mixed Models for Repeated Measures Should Include Time-by-Covariate Interactions to Assure Power Gains and Robustness Against Dropout Bias Relative to Complete-Case ANCOVA. Ther Innov Regul Sci. 2022 Jan;56(1):145-154. doi: 10.1007/s43441-021-00348-y. Epub 2021 Oct 21. PMID: 34674187.